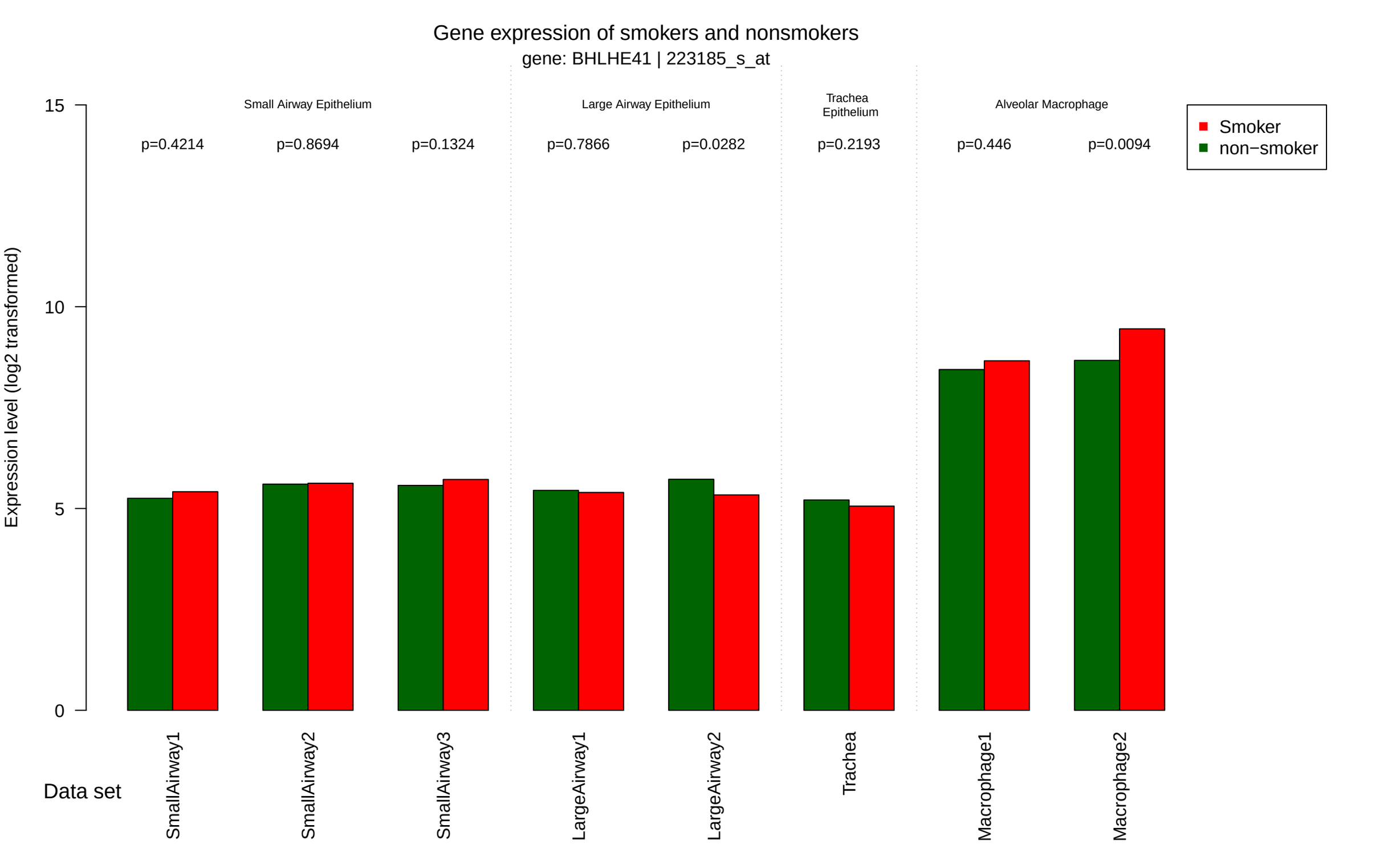
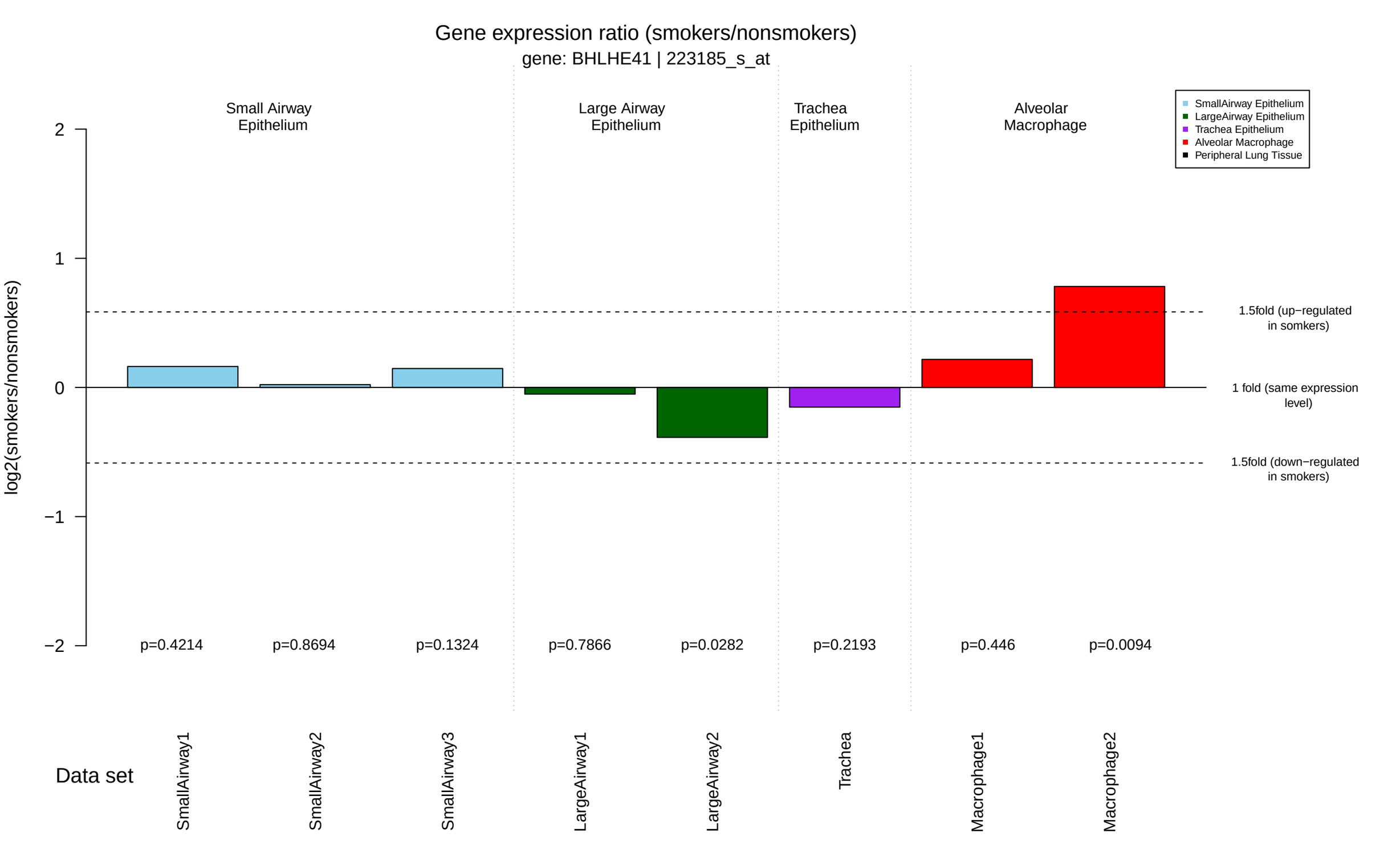
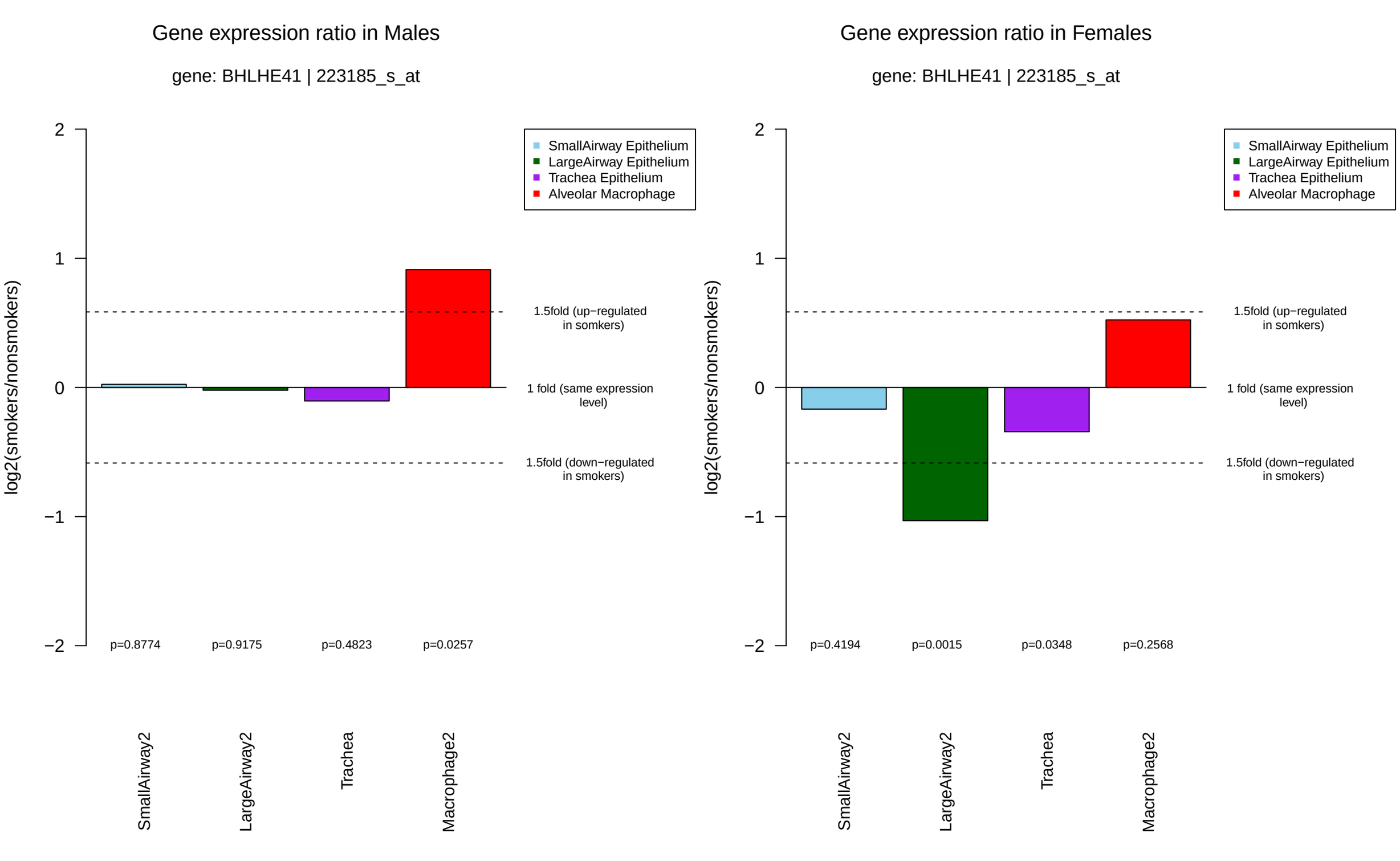
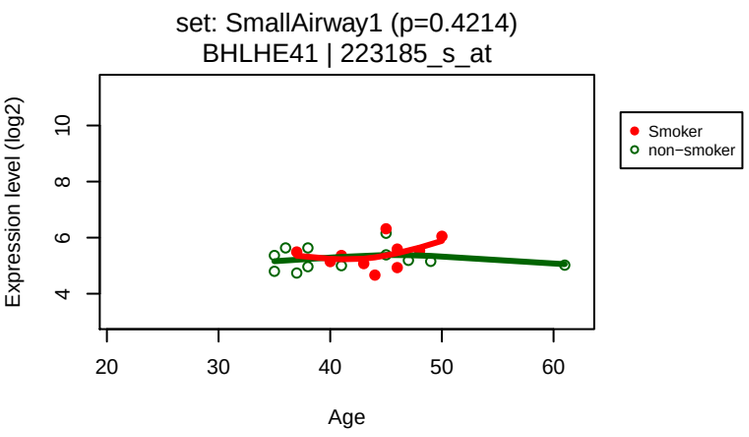
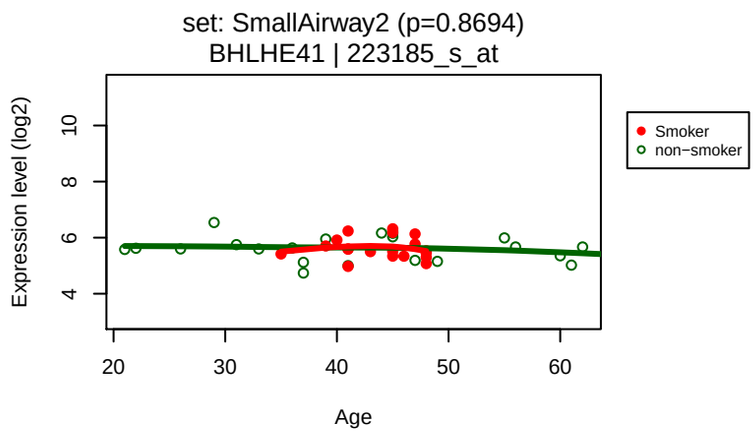
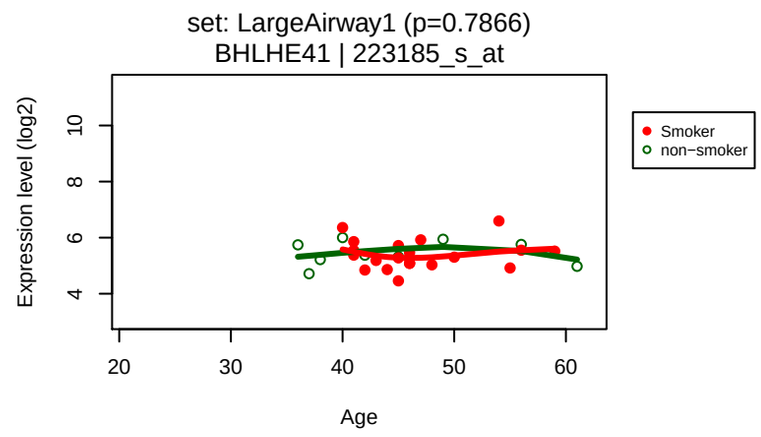

Gene expression of smokers and nonsmokers
gene: BHLHE41 | 223185_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4214
p=0.8694
p=0.1324
p=0.7866
p=0.0282
p=0.2193
p=0.446
p=0.0094
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

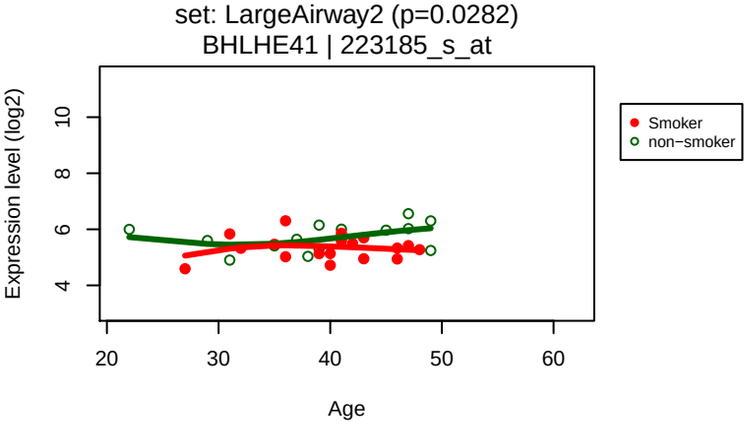

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE41 | 223185_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4214
p=0.8694
p=0.1324
p=0.7866
p=0.0282
p=0.2193
p=0.446
p=0.0094
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

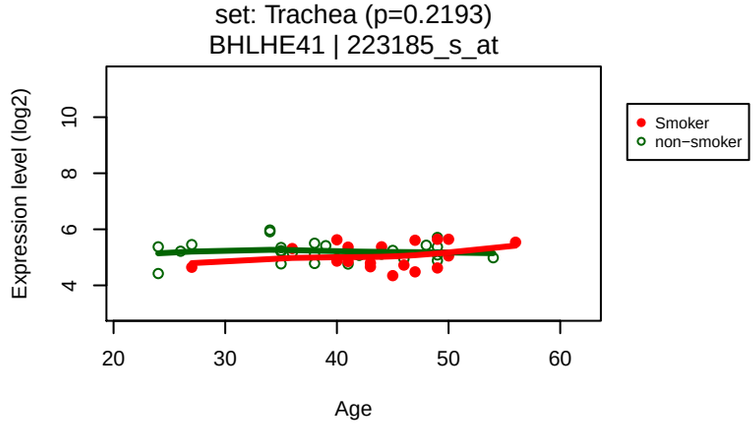

Gene expression ratio in Males
gene: BHLHE41 | 223185_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8774
p=0.9175
p=0.4823
p=0.0257
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE41 | 223185_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4194
p=0.0015
p=0.0348
p=0.2568
SmallAirway2
LargeAirway2
Trachea
Macrophage2

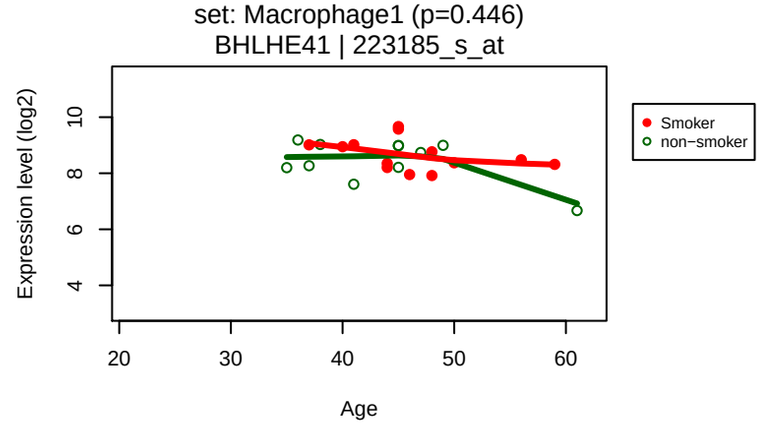

set: SmallAirway1 (p=0.4214)
BHLHE41 | 223185_s_at
Expression level (log2)
Age
Smoker
non−smoker

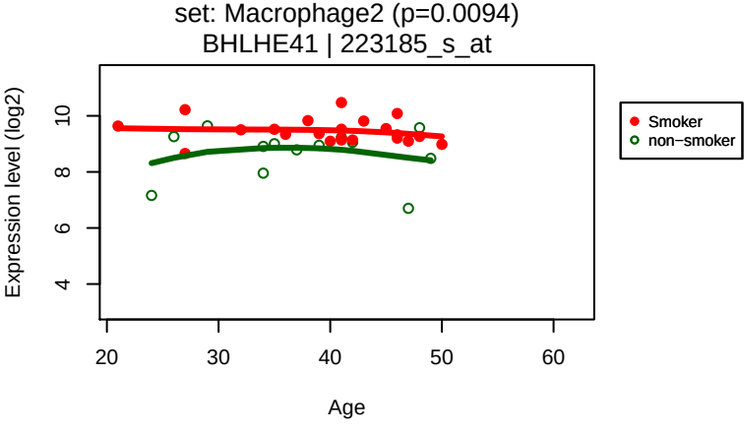

set: SmallAirway2 (p=0.8694)
BHLHE41 | 223185_s_at
Expression level (log2)
Age
Smoker
non−smoker

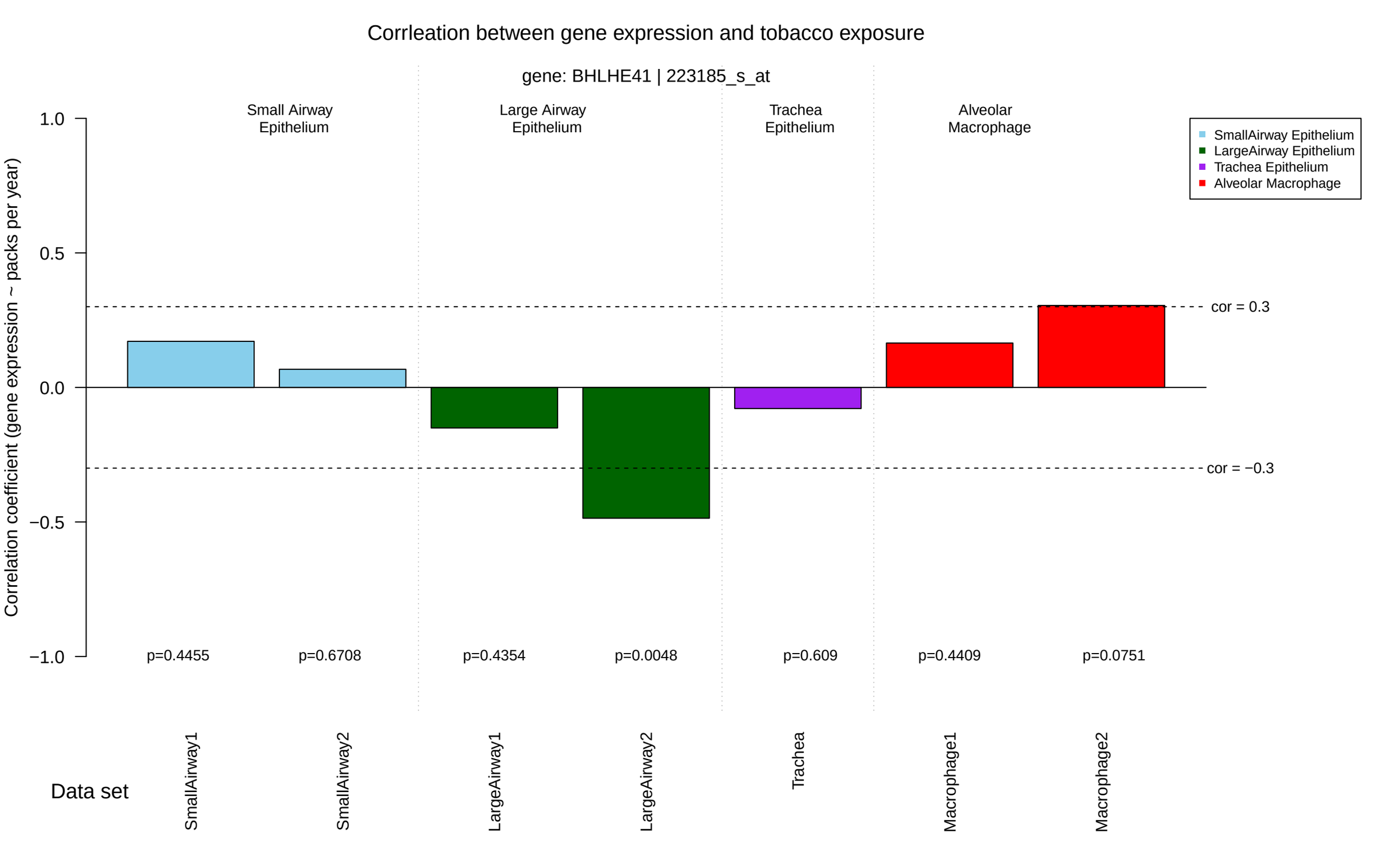

set: LargeAirway1 (p=0.7866)
BHLHE41 | 223185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0282)
BHLHE41 | 223185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2193)
BHLHE41 | 223185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.446)
BHLHE41 | 223185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0094)
BHLHE41 | 223185_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE41 | 223185_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4455
p=0.6708
p=0.4354
p=0.0048
p=0.609
p=0.4409
p=0.0751
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2